POONAM SIKARWAR

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RESEARCH SUMMARY

Materials scientist and electrochemist with expertise in 3D and low dimensional halide perovskites, photoelectrochemical water splitting, light emitting materials, and silicon crystal growth processes. Research interests include interfacial charge transfer, defect engineering, ML-assisted material discovery, and structure-property relationships in functional materials. Experienced in materials synthesis, device fabrication, advanced spectroscopy, electrochemical characterization, and silicon loss of structure analysis, with a strong focus on developing sustainable energy conversion and simulation expertise.

EDUCATION

Direct Ph.D. (MS + Ph.D.) - Chemical Engineering (CGPA: 8.07)

Indian Institute of Technology Madras, India | July 2017 – Mar 2024

Thesis title: Design of Semiconducting Perovskite Halides and their Application in Optoelectronic Devices.
Thesis advisors: Prof Aravind Kumar Chandiran

Bachelor of Technology - Chemical Engineering (CGPA: 8.38)

Institute of Technology & Management, India | Jul 2010 – May 2014

Dissertation title: Lemongrass oil extraction through hydro-distillation and microwave assisted distillation and its physiochemical property analysis.

RESEARCH EXPERIENCE

Quantitative Silicon Researcher, Ingot and Wafer Technology, Reliance New Energy Limited

Jun 2023 – Present | Jamnagar/Navi Mumbai, India

- Managed R&D operations for semiconductor ingot and wafer manufacturing, bridging the gap between pilot-scale research and production-level facilities.
- Worked on the production floor of a live Czochralski crystal growth fab, building mathematical models to determine crystal length meeting target resistivity ($\leq 1.5\ \Omega\cdot cm$) as a function of melt dopant concentration.
- Developed Python/FlexSim digital twins to simulate manufacturing workflows on the shop floor, identify bottlenecks, and evaluate scale-up strategies, supporting Capex reduction and AI-readiness initiatives.
- Implemented wafer metrology and in-line inspection plans on the fab floor; partnered with cross-functional production teams to deliver a 10% yield-loss reduction and align simulation outputs with production timelines, ensuring seamless integration in a complex research environment.
- Designed 100+ hot-zone components in AutoCAD for installation on production crystal pullers, optimizing single-crystal formation and thermal uniformity.
- Developed a machine-learning predictive model for loss-of-structure (LOS) detection in Czochralski (Cz) silicon crystal growth, combining in-situ process signals with Python-based classification algorithms for real-time quality monitoring; directly transferable to any operando system.
- Acquired in-depth understanding of Cz-crystal growth thermodynamics, melt-solid interface dynamics, and crystalline defect formation mechanisms - foundational to understanding charge carrier trapping at defect sites, thin film nucleation and growth mechanism in energy materials.

Doctoral Assistant, Group of Prof Aravind Kumar Chandiran at Indian Institute of Technology Madras

July 2017 – Mar 2024 | Chennai, India

- Synthesised and characterised a library of lead-free halide double perovskite compositions ($Cs_2AgIn_xBi_{1-x}Cl_6$, $Cs_2AgIn_xSb_{1-x}Cl_6$, $Cs_2AgBi_xSb_{1-x}Cl_6$) via hydrothermal synthesis and established a universal structural-optoelectronic property correlation driving, 144× photoluminescence enhancement and ~1 μs excited-state lifetime through mixed trivalent cation engineering of $[AgCl_6]^{5-}$ octahedral distortion. Further Established electric-field-driven structural distortions in $Cs_2NaIn_xBi_{1-x}Cl_6$ double perovskites using FTIR spectroscopy, correlating octahedral distortions with enhanced photoluminescence behaviour leading to a publication in *Commun Mater (Nature) and ChemPhysChem.*
- Achieved photoelectrochemical water-oxidation efficiency close to 70% of its theoretical limit and also designed a homemade “Photoelectrochemical cell” for solar water splitting applications. (*published in Physical Review Applied*).

- Chemically tuned the dimensionality in halide perovskites by A-site engineering, that provided an opportunity to obtain the properties desired for optoelectronic devices and reported in *Inorganic Chemistry*.
- Developed a broadband light-emitting organic-inorganic hybrid zero-dimensional non-centrosymmetric material where the non-centrosymmetric nature of the material is confirmed using piezo-force imaging and polarization-electric field (P-E) loop. The material showed a broad emission with photoluminescence (PL) quantum yield of 13.1 %. *(manuscript submitted to Applied Physics Letters)*
- Designed, developed, and validated a single-semiconductor white-light emitting material based on trivalent mixed halide double perovskites ($Cs_2AgBi_{1-x}In_xCl_6$), demonstrating broadband emission from 400–750 nm with carrier lifetimes of around 990 ns across the composition range, directly addressing the LED phosphor challenge of achieving full visible spectrum from a single material, resulting in a granted Indian patent *(Patent No. 573358, 2025).*
- Demonstrated fabrication of white-light emitting devices by dip-coating the emissive perovskite compound with PMMA polymer onto transparent substrates, achieving visible white light output upon UV excitation, establishing a proof-of-concept device pathway from material to light-emitting application.
- Synthesized and characterized a wide range of functional materials, including low-dimensional (0D-2D) halide perovskites, transition-metal complexes, ferroelectric materials, and metal oxides, targeting enhanced stability, optoelectronic performance, and catalytic activity
- Applied advanced structural, optical, and electrochemical characterization techniques, including single-crystal and powder XRD, UV-Vis, photoluminescence, FTIR, Raman, XPS, EIS, and dielectric spectroscopy, to establish structure-property relationships in functional energy materials.
- Mentored junior researchers in perovskite synthesis, structural characterization, and photoelectrochemical measurements, contributing to multiple collaborative peer-reviewed publications.

Junior Research Fellow, Group of Prof Prasenjit Mondal at Indian Institute of Technology Roorkee
Dec 2015 – Dec 2016 | Roorkee, India

- Conducted research on "Adsorption of arsenic and fluoride from ground water onto laterite soil and its stabilization bearing spent adsorbent in clay bricks".

Industrial Internship (F&B Sector) at Mondetez International-Cadbury, Madhya Pradesh
Jun 2013 – Jul 2013 | Malanpur, India

- "Quality Control" in new product development.

Industrial Internship (Cement Plant) at Heidelberg Cement, Madhya Pradesh
Jul 2013 – Aug 2013 | Jabalpur, India

- Quality control of "Diamond Cement Properties".

PATENT

Granted

1. White light emission from a single semiconductor material based on trivalent mixed halide perovskite. A. Tamilselvan, **P. Sikarwar,** A.K. Chandiran*. Indian patent number: 493546. Issued Feb, 2024*

PUBLICATIONS (PEER-REVIEWED)

8 publications (6 published, 2 under review) | 6 first-author or shared-first-author

Published

1. **P. Sikarwar**, P. Siwach, N.V.P. Chandra, P.K.S. Antharjanam, A.K. Chandiran. "Dimensional reduction of Cs2AgBiBr6 using alkylammonium cations - CH3(CH2)nNH3+ (n = 1, 2, 3, and 6) and their role in structural and optoelectronic properties.", Inorganic Chemistry, 2023. DOI: https://doi.org/10.1021/acs.inorgchem.3c00571
2. P. Siwach, **P. Sikarwar**, J.S. Halpati, and A.K. Chandiran. "Design of above-room-temperature ferroelectric two-dimensional layered halide perovskites.", Journal of Material Chemistry A, 2022. *(Shared first author).* DOI: 10.1039/d1ta09537d
3. **P. Sikarwar**, K.T. Indraja, A. Tamilselvan, A.K. Chandiran. "Highly Efficient Photoelectrochemical Water Oxidation Using Cs2AgMCl6 (M- In, Bi, Sb) Halide Double Perovskites.", Physical Review Applied, 2023. DOI: https://doi.org/10.1103/PhysRevApplied.19.044083
4. A. Tamilselvan, R. Kashikar, **P. Sikarwar**, S. Antharjanam, BRK. Nanda, and A.K. Chandiran, "Manipulation of parity and polarization through structural distortion in light-emitting halide double perovskites.", Commun Mater (Nature), 2021. DOI: https://doi.org/10.1038/s43246-021-00172-9
5. **P. Sikarwar**, A. Anand, A.K. Chandiran. "Octahedral distortion induced photoluminescence in Cs2NaInxBi1-xCl6 halide double perovskites.", ChemPhysChem, 2025. DOI: https://doi.org/10.1002/cphc.202401036

6. P. Siwach, **P. Sikarwar**, S.A. Rajput, P.K.S. Antharjanam and A.K. Chandiran. "The effect of halogenated spacer cation on structural symmetry breaking in 2D halide double perovskites.", Chem. Commun., 2022. DOI: 10.1039/d2cc02747j

Under Review

7. P. Siwach; **P. Sikarwar**; M. Gupta; A.K. Samuel; P.K.S. Antharjanam; B.R.K. Nanda; A.K. Chandiran. Dimensional tuning in halide double perovskites using aliphatic and aromatic organic spacers. *(Shared first author) (Journal of Solid-State Chemistry)*
8. **Sikarwar, P.**; Siwach, P.; Rajput, S.; Anand, A.; Raj, V.; Antharjanam, S.; Chandiran, A. K. Non-centro-symmetric antimony-based organic-inorganic hybrid halides for broadband white light emission with 13.1% room temperature photoluminescence quantum yield. *(Applied Physics Letters)*

TECHNICAL SKILLS

- **Synthesis & Fabrication:** Single crystals, nanostructured powders, thin films | Hydrothermal, sol-gel, solid-state, co-precipitation, evaporation with slow-cooling, evaporation with fast-cooling, seed crystallization | Spin coating, doctor blading, electrochemical deposition, electrophoretic, spray coating and chemical bath deposition | Screen printing | CVD | PVD
- **Structural & Morphological Characterization:** SCXRD, PXRD, AFM, PFM, SECM
- **Spectroscopy:** XPS, FTIR, Raman | UV-Vis, PL, TRPL, confocal fluorescence | In-situ temperature and voltage dependent PL | TGA, DSC
- **Elemental & Compositional:** EDS, ICP-MS/OES, GC-MS
- **Electrochemical & Dielectric:** EIS, CV, LSV, GCD/PCD, dielectric spectroscopy
- **Crystal Growth:** Czochralski (Cz) silicon crystal growth process and hot-zone design
- **Computation & Data:** Python (NumPy, Pandas, Matplotlib, SciPy, Plotly, Streamlit) | ML (Scikit-learn, TensorFlow, and PyTorch) | MATLAB | VESTA, CASA XPS, Z-View, Mendeley, EC-Lab | AutoCAD | FlexSim (digital Twin) | Origin, SQL, Tableau, Power BI | SolidWorks, MS Office.
- Scientific writing, technical reporting, patent drafting, proposal preparation, and oral presentation.

TEACHING EXPERIENCE

Teaching Assistant, Department of Chemical Engineering, Indian Institute of Technology Madras
- Introduction to Research, Jul – Dec 2018
- Post graduate course on Photocatalysis, Jul – Dec 2019
- B. Tech Lab on Experimental Thermodynamics, Jan – May in 2019, 2020 and 2022
- Material Science for Chemical Engineers, Jan – May 2021
- AICTE-sponsored Short-Term Training Programme on Chemical Engineering Thermodynamics Lab for faculties from different academic institutions, June 2019
- Confocal Laser Scanning Microscope, Jul – Nov 2020

PEER REVIEWER

- For "Wiley Laser & Photonics Reviews"

CONFERENCES

- Shared my research work in an online international conference entitled "Photoelectrochemical Water Oxidation using Halide Double Perovskites" at 2022-MRS Spring Meeting and Exhibit. (may 2022)
- Poster presentation entitled "Organic Cation Engineering and its Effect on Distortion in 2D Halide Perovskites" in 2D-HAPES 2021 International conference, Spain. (Nov 2021)
- Poster presentation entitled "Photoelectrochemical water oxidation using Halide double Perovskites" Conference (MATSUS Spring 23) at Valencia, Spain. (Mar 2023)
- Paper presentation entitled "Cs2AgMCl6 (M= In3+, Bi3+ or Sb3+) halide perovskites for solar water splitting" in the CHEM+ 2020 at IIT Madras. (Feb 2020)

- Attended “DAE SSPS 2019 symposium” at IIT Jodhpur. (Dec 2019)
- Attended SEMICON India 2025, at Yashobhoomi (IICC), Dwarka, New Delhi. (Sep 2025)
- Poster presentation entitled “Halide Perovskite for Light Emitting Applications” in Envision 2022, organized by IITM Research Park. (Nov 2022)

TRAINING & CERTIFICATIONS

- Single Crystal XRD Data Analysis at Sophisticated Analytical Instruments Facility, IIT Madras. (Jul – Aug 2019)
- Six Sigma Yellow Belt

PROJECTS

AI Metrology Pipeline for Semiconductor Wafer Defect Classification:

PyTorch, OpenCV, Grad-CAM, Scikit-Learn, Pandas

- Engineered an end-to-end Convolutional Neural Network (CNN) to automate defect spatial signature classification (Center, Edge-Ring, Scratch, etc.) across 811,000+ Wafer Bin Maps (WM-811K dataset), targeting early detection of equipment drift and reducing scrap wafer rates.
- Overcame severe 5,275:1 class imbalance between baseline wafers and rare physical anomalies by implementing strategic under-sampling and dynamically weighted Cross-Entropy Loss, achieving a 94.2% overall accuracy and 0.91 macro F1-score.
- Integrated Grad-CAM (Gradient-weighted Class Activation Mapping) to extract spatial feature weights from the final convolutional layer, translating "black-box" neural network outputs into interpretable heatmaps for process engineers to perform physical root-cause analysis.
- Built robust data preprocessing pipelines to normalize diverse spatial array dimensions (26*26 to 52*52) into uniform tensors, ensuring stable gradient descent and preventing computational bottlenecks during training.

Project: Real-Time Yield & RCA Diagnostic Tool (Digital Kaizen)

- Led a Digital Kaizen initiative to automate the analysis of silicon ingot Loss of Structure (LOS) events across multiple crystal pulling machines.
- Built a local HTML/JS web application that parses raw operator data to instantly visualize the distribution of Crown, Body, and Tail LOS events by specific length categories (e.g., 0-500mm, 501-1000mm).
- Implemented automated cross-cycle comparison metrics, allowing engineers to benchmark trial runs against standard operating procedures (SOPs) based on dynamically calculated date ranges.
- Impact: Significantly accelerated the RCA process by providing immediate visual indicators distinguishing between localized equipment issues (e.g., bad heater) and systemic process failures

ACHIEVEMENTS

- Participated and won best “Paper Presentation” in CHEM+ 2020 entitled: "Cs2AgMCl6 (M= In3+, Bi3+ or Sb3+) halide perovskites for solar water splitting” at IIT Madras, Feb 2020
- Awarded among top ten ideas in “New Generation Ideation Contest” organized by “HPCL GREEN R&D”, Bangalore, India. The proposed idea was to use the solar panels for sea water desalination and parallely improve their efficiency, Sep 2019

CO-CURRICULAR ACTIVITIES

- Member of organising committee “Energy Summit 2022” organized by the Energy Consortium at IITM Research Park. (Dec 2022)
- Member of organizing committee for “ChemSymp 2018, IIT Madras, Mar 2018